Cong Hoan Ngo

Software Engineer

Hanoi, Vietnam

conghoan.0406@gmail.com | hwanngo.github.io | github.com/hwanngo | linkedin.com/in/hwanngo

Summary

Software Engineer with 7+ years building backend services, REST APIs, and web applications for US and Singapore businesses. Builds Python and Java backends (Flask, Spring Boot), including a move from Flask servers to serverless AWS (Lambda, API Gateway, S3) that cut infrastructure cost by about 25%, and web front ends in Angular and React. Since 2023 also builds LLM-powered pipelines and tools, and data platforms with Apache Airflow and dbt on Snowflake and BigQuery, with infrastructure as code (OpenTofu) and workloads on Kubernetes.

Skills

Programming: Python, SQL, Java, JavaScript/TypeScript

Software Engineering: REST APIs, FastAPI, Flask, Spring Boot, React, Angular

Cloud & Infrastructure: AWS, GCP, Azure, OpenTofu/Terraform, Docker, Kubernetes, Helm, CI/CD

AI & LLM: OpenAI & Anthropic APIs, prompt engineering, LLM evaluation, embeddings & pgvector

Databases & Warehousing: Snowflake, BigQuery, PostgreSQL, MySQL, SAP HANA, Redis

Data Engineering: ETL/ELT, Apache Airflow, Airbyte, dbt, data modeling, data quality, web scraping, Selenium, Crawl4AI

BI & Analytics: Metabase, Tableau

Languages: English - C2 Proficient (CEFR)

Work Experience

Pod Co.

Data Engineer (Data & Software Engineering) · Apr 2025 – Present · Remote · US-based company

- Built LLM-powered discovery on scraped social media: 100+ potential brands per week, over 30% became partnerships.
- Build internal web apps and tools for the team with React, such as a brand detection tool.
- Prototyped an LLM text-to-SQL analytics assistant on Snowflake with query guardrails and LLM evaluation.
- Maintain infrastructure as code with OpenTofu (Terraform) and monitor data workloads on Kubernetes (Helm).
- Build and operate 150+ Airflow DAGs and Airbyte pipelines into Snowflake for Wholesale, B2B, and Distribution.
- Develop dbt models and tests on Snowflake behind 200+ scheduled Metabase reports for business, finance, and sales.
- Made dashboards faster to refresh and less prone to breaking, holding cost flat and cutting it 20–30% on some workloads.

VMO Holdings Technology JSC

Data Engineer · Dec 2023 – Apr 2025 · Vietnam

- Migrated legacy cron pipelines to Apache Airflow, improving scheduling, monitoring, and maintainability.
- Built Python ETL workflows for multi-source healthcare and pharma data in a data lake and warehouse (Azure, BigQuery).
- Supported critical data source migrations while preserving data integrity and downstream compatibility.
- Improved pipeline reliability and reconciliation, enabling analytics delivery for 200+ recurring reports.
- Investigated data discrepancies with analytics teams and delivered Tableau dashboards for business users.

Maxflow R&D Co. Ltd.

Software Engineer / Full-stack Developer · Jan 2021 – Dec 2023 · Vietnam

- Moved Python Flask services to serverless AWS (Lambda, API Gateway, S3), cutting infrastructure cost by about 25%.
- Developed core features and REST APIs for an industrial simulation and analytics platform with configurable workflows.
- Implemented simulation modules for distance, energy, wear, and lifecycle cost across 100+ modeled scenarios.
- Built data preparation and feature extraction pipelines on Snowflake for predictive maintenance experiments.

FPT Information System Corp.

Back-end Developer · Dec 2019 – Dec 2020 · Vietnam

- Built Java (Spring Boot) backend services for a national public procurement platform in a 5-engineer team.
- Contributed to Angular-based front-end improvements, reducing average page load time by approximately 20%.

House3D Co. Ltd.

Web Developer · Feb 2019 – Dec 2019 · Vietnam

- Developed and maintained customer-facing web features for an interior design technology platform.

Framgia Inc. / Sun* Inc.

Web Developer Intern · Jun 2018 – Sep 2018 · Vietnam

- Built a Ruby on Rails e-commerce project in a 3-member team with PayPal, BaoKim, and NganLuong payments.

Projects

Discode: AI Coding Assistants in Discord

Personal Project · Python, PostgreSQL, Redis, Docker

- Built a Discord bot platform that runs AI coding CLIs (Claude, Codex, OpenCode) as stateful, thread-based chat sessions.
- Designed a multi-process bot/runner/dispatcher on PostgreSQL and Redis with an outbox and idempotent job claims.

Chatty Buddy: Multi-Provider AI Chat Client

Personal Project · React, TypeScript, Zustand, Vite

- Built a PWA chat client for OpenAI-compatible, Anthropic, and Ollama APIs in React 19, TypeScript, and Zustand.
- Implemented streaming adapters, tool permissions, and versioned backups; CI runs type checks, Vitest, and audits.

Education

VNU – Hanoi University of Science

Bachelor of Science in Computer and Information Science · 2016 – 2021 · Vietnam

- Coursework focus: Software Engineering, Database Systems, Data Structures and Algorithms.

Certifications

IBM Data Architecture · May 2025

Google Associate Cloud Engineer · Dec 2024

Google Advanced Data Analytics · Nov 2024

IBM Data Engineering · Jul 2024

Microsoft Azure Data Engineering Associate · Jun 2024

AWS Cloud Solutions Architect · Jun 2024

Honors & Awards

Employee of the Year (VMO Year-End Award, VMO Holdings) · 2024